Samarth Singh Rawat

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Summary

Applied AI developer and B.Tech IT student at KIIT Bhubaneswar, specializing in interactive systems, behavioral architectures, and AI-enhanced user experiences. Experienced in integrating embedded systems, real-time interfaces, and intelligent workflow automation into cohesive, product-oriented applications that merge machine learning, IoT, and scalable web solutions.

Education

Kalinga Institute of Industrial Technology (KIIT), Bhubaneswar, Odisha
Bachelor of Technology in Information Technology — 2026
8th Semester CGPA: 8.24

Lal Bahadur Shastri Public School, Kota
12th Grade, CBSE — 2022
Percentage: 79.4%

GMR DAV Varalakshmi Public School, Dhenkanal
10th Grade, CBSE — 2020
Percentage: 90%

Fields of Interest

LLMs, Artificial Intelligence, Fine tuning, Deep Learning, Agentic AI, Data Visualization, Machine Learning, IoT Systems, Embedded Systems, Human-Computer Interaction, Data Analysis, Web Designing

Skills

Languages	Python, JavaScript, C, HTML, CSS, ABAP
AI / ML	LangChain, LangGraph, HuggingFace, Hugging Face Transformers, RAG, Groq LPU, Llama 3.3, Gemini, Whisper, TensorFlow, Scikit-learn, Agentic AI, Generative AI, Machine Learning
Web Development	FastAPI, Pydantic, REST APIs, SQLite3, React.js, Tailwind CSS, Vite, Framer Motion, Recharts
Cloud & Tools	Git, VS Code, Vercel, Render, PythonAnywhere, Figma, Arduino, Power BI, LaTeX, Microsoft Office
Core CS	OOPs, DBMS, Data Structures, Operating Systems, Data Science

Projects / Open-Source

1. Critic-OS (Completed)

Python (Flask), Groq (Llama-3.3), HuggingFace, Redis, Spotify API, Last.fm API, Vercel

- Developed an AI-driven web application that generates real-time, satirical music critiques using Groq (Llama-3.3) and 6 distinct AI personas (e.g., Gordon Ramsay, Cyberpunk Hacker).
- Engineered a parallel enrichment engine that integrates Spotify, Last.fm, and LRC_LIB APIs with HuggingFace emotion classification to analyze user playlists in under 6 seconds.
- Optimized for production using Vercel Serverless and Redis-backed session management to ensure high-speed performance and efficient memory usage.

2. Movie Recommendation System (Completed)

Python, Scikit-learn, NumPy, HTML

- Built a collaborative filtering engine to generate personalized movie suggestions.
- Processed large datasets and designed a simple web UI for user input and recommendations.

3. IoT Smart Alarm Clock (Completed)
ESP32-C3, C++, LittleFS, I2C/SPI, FreeRTOS, Python (Asset Packing)

- **Memory-Constrained Architecture**: Engineered a custom `.bin` streaming animation engine to bypass ESP32-C3 RAM limitations, enabling high-quality animations via LittleFS frame-streaming with 16-byte header verification.
- **Resource Optimization**: Implemented a multi-tier throttling strategy (I2C capping, sensor polling reduction with median filtering, and UI string caching), reducing CPU overhead by 40% and enhancing system stability.
- **Modular Firmware Design**: Developed a feature-toggle architecture that reduced BSS memory allocation by 65%, ensuring compatibility with WiFi/BT stacks while maintaining a responsive, interrupt-driven multi-page menu system.
- **System Reliability**: Integrated a heterogeneous sensor suite (PIR, Ultrasonic, IMU) for gesture control and motion wake, synchronized with a dual-source (NTP + Hardware RTC) timekeeping system for offline resilience.

4. ISL Hand Gesture Recognition Using Random Forest (Completed)
Python, Scikit-learn, NumPy, Pandas

- Processed ISL hand gesture datasets in CSV format with landmark and movement data.
- Trained a Random Forest model to classify Indian Sign Language gestures accurately.

Strengths

- Systems-oriented embedded developer experienced in modular ESP32 architecture, real-time interaction systems, and hardware-software integration.
- Strong problem-solving and systems-thinking skills with focus on scalable, maintainable, and extensible design patterns.
- Experienced in integrating sensors, displays, RTC modules, animation systems, and input handling into cohesive embedded applications.
- Rapid prototyping and iterative development capability with practical debugging experience across complex firmware and hardware interactions.
- Cross-domain technical exposure spanning embedded systems, UI/UX interaction design, simulation workflows, and AI-assisted development tools.

Hobbies

Reading books, Watching movies, Learning new things, Building and innovating, Listening to stories and podcasts